

Jupiter Database

Technical Reference & Exploratory Analysis

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Note: This is a living reference document. Update it as you learn more. It serves as a day-to-day cheat sheet while building the system, and as raw material for the Data chapter of the thesis report.

1. Database Overview

JupiterXL is Denmark's national database for groundwater and drinking water chemistry, maintained by GEUS. The version used is PCJupiterXL Plus, downloaded 26 March 2026, restored into PostgreSQL 16 with PostGIS 3.4.

Property	Value
Database format	PostgreSQL 16 custom backup (.backup)
Compressed / uncompressed size	3.7 GB / 19.9 GB
Schema name	pcjupiterxlplusviews
Total tables	276 (incl. ~170 code_* lookup tables)
Boreholes	477,105
Drinking water plants (drwplant)	103,195 total; 69,851 active public
Groundwater chemistry samples	508,413
Groundwater chemistry analyses	14,671,938
Chemical compounds tracked	3,026
Compounds with consumer limit defined	1,217
WSA polygons (Schullehner 2022)	3,694 (loaded as pcjupiterxlplusviews.wsa)
WSA-Plant linkage rows	7,422 (loaded as pcjupiterxlplusviews.wsa_plant)
Data freshness	Nightly export — as recent as March 2026

2. Thesis Scope & Primary Question Types

The system is designed around two core citizen-facing question types. Groundwater monitoring is secondary.

Question type	Primary tables	Notes
"Where does my drinking water come from?"	wsa, wsa_plant, drwplant, drwplantintake, borehole	Point-in-polygon on WSA polygons gives plantid(s). drwplant gives name, address, watertype.
"Is my water safe / has PFAS been detected?"	wsa, wsa_plant, pltchemsample, pltchemanalysis, compoundlist, limitlist	Same spatial lookup then plt* chemistry chain with quality filters and limit comparison.
"Nitrate in groundwater near me?" (secondary)	borehole, grwchemsample, grwchemanalysis, compoundlist, limitlist	grw* chain. Use borhtownno2007 + code_808 for municipality filtering.

3. System Architecture & Join Chains

3.1 Core spatial lookup

The WSA dataset (Schullehner 2022) is the critical bridge. Jupiter has no supply zone concept. All coordinates are EPSG:25832 (UTM32 ETRS89).

```
SELECT wp.plantid, p.plantname, p.plantaddress, p.watertype
FROM pcjupiterxlplusviews.wsa w
JOIN pcjupiterxlplusviews.wsa_plant wp ON wp.wsaid = w.wsaid
JOIN pcjupiterxlplusviews.drwplant p ON p.plantid = wp.plantid
WHERE ST_Contains(w.geom, ST_SetSRID(ST_MakePoint(:x_utm32,
:y_utm32), 25832))
AND w.end = 2019;
```

Note: User addresses must be geocoded and converted to UTM32 before spatial lookup. WSA coverage ends 2019 -- disclose this limitation to users.

3.2 Full drinking water safety query (verified end-to-end)

Proven with real data: Copenhagen city centre (725337, 6176210) resolves to HOFOR Soendersoe and Slangerup plants, nitrate 1.3-2.6 mg/l, well within the 50 mg/l limit.

```
WITH my_plants AS (
  SELECT wp.plantid
  FROM pcjupiterxlplusviews.wsa w
  JOIN pcjupiterxlplusviews.wsa_plant wp ON wp.wsaid = w.wsaid
  WHERE ST_Contains(w.geom, ST_SetSRID(ST_MakePoint(:x, :y),
25832))
  AND w.end = 2019
)
SELECT p.plantname, p.plantaddress, p.watertype,
c.long_text AS compound, s.sampledate, a.amount,
```

```

    u.longtext AS unit, l.consumer_max AS legal_limit,
CASE
    WHEN a.attribute IS NOT NULL THEN 'NOT DETECTED'
    WHEN l.consumer_max IS NOT NULL AND a.amount > l.consumer_max
THEN 'EXCEEDS LIMIT'
    ELSE 'OK'
END AS status
FROM my_plants mp
JOIN pcjupiterxlplusviews.drwplant p ON p.plantid = mp.plantid
JOIN pcjupiterxlplusviews.pltchemsample s ON s.plantid = p.plantid
JOIN pcjupiterxlplusviews.pltchemanalysis a ON a.sampleid =
s.sampleid
JOIN pcjupiterxlplusviews.compoundlist c ON c.compoundno =
a.compoundno
JOIN pcjupiterxlplusviews.code_752 u ON u.code = a.unit
LEFT JOIN pcjupiterxlplusviews.limitlist l ON l.compoundno =
a.compoundno
WHERE a.compoundno = :compoundno
AND (a.qualitycontrol IS NULL OR a.qualitycontrol NOT IN
(4,5,6,8,12,13,14,15))
AND (s.samplestatus IS NULL OR s.samplestatus IN (2,4,6,8,10,14))
AND (s.project IS NULL OR s.project = 'DRV')
AND s.sampledate <= NOW()
ORDER BY s.sampledate DESC LIMIT 20;

```

3.3 Compound lookup

```

SELECT compoundno, long_text, short_text, casno, grwunit
FROM pcjupiterxlplusviews.compoundlist
WHERE long_text ILIKE '%nitrat%' OR short_text ILIKE '%nitrat%';

```

3.4 PFAS group query

```

SELECT c.compoundno, c.long_text
FROM pcjupiterxlplusviews.compoundlist c
JOIN pcjupiterxlplusviews.compoundgroup cg ON cg.compoundno =
c.compoundno
WHERE cg.compoundgroupno = 110;

-- Use in chemistry query:
WHERE a.compoundno IN (
    SELECT c.compoundno FROM pcjupiterxlplusviews.compoundlist c
    JOIN pcjupiterxlplusviews.compoundgroup cg ON cg.compoundno =
c.compoundno
    WHERE cg.compoundgroupno = 110)

```

4. Key Tables

4.1 Spatial tables (external datasets)

Table	Source	Rows	Description
wsa	Schullehner 2022	3,694	Water Supply Area polygons, EPSG:25832. Columns: wsa_id, start, end, geom (MULTIPOLYGON).
wsa_plant	Schullehner 2022	7,422	Links WSAID to Jupiter PLANTID with temporal coverage (start_year, end_year). One zone can have multiple plants.

4.2 Drinking water tables

Table	Key columns	Description
drwplant	plantid, plantname, plantaddress, municipalityno2007, active, dataowner, watertype, xutm32euref89, yutm32euref89	Central drinking water plant table. 103,195 rows. Filter: active!=2, dataowner=KO for public plants.
drwplantintake	plantid, boreholeid, intakeid, intakeusage, startdate, enddate	Links plants to source boreholes. Use startdate/enddate for temporal accuracy. intakeusage codetype=816.
drwplantcompanytype	plantid, companytype	V01+V02 = public municipal waterworks (almene vandvaerker). V95 and V40 are the two most common other types.
pltchemsample	sampleid, plantid, sampleddate, samplestatus, qualitycontrol, project, pipesite, pipeaddress	Drinking water samples at plant or distribution network. Apply quality filters.
pltchemanalysis	analysisid, sampleid, compoundno, amount, unit, attribute, qualitycontrol, detectionlimit	Drinking water chemistry. One row per compound per sample. Check attribute before flagging exceedance.

4.3 Groundwater tables

Table	Key columns	Description
borehole	boreholeid, boreholeno, latitude, longitude, xutm32euref89, yutm32euref89, borhtownno2007, use, status, drilldepth	477,105 boreholes. Use borhtownno2007+code_808 for municipality, not raw municipal text field.
intake	intakeid, boreholeid, intakeno	Depth interval within borehole. First, an intake is created on a borehole. Then a screen (filter) is attached to it.
screen	screenid, intakeid	Physical filter element. Determines which depth the water comes from.

Table	Key columns	Description
grwchemsample	sampleid, boreholeid, intakeid, sampleddate, samplestatus, qualitycontrol	508,413 rows. Has both boreholeid and intakeid -- intake join optional for basic queries.
grwchemanalysis	analysisid, sampleid, compoundno, amount, unit, attribute, detectionlimit, qualitycontrol	14.6M rows. Apply same quality filters as plt*. Always filter by compoundno + date range first.

4.4 Compound & limit tables

Table	Key columns	Description
compoundlist	compoundno, long_text, short_text, casno, grwunit, drwunit	3,026 compounds. Names in Danish. long_text is the primary lookup field.
compoundgrouplist	compoundgroupno, longtext, shorttext	Group definitions. compoundgroupno=110 = Perfluorerede stoffer (PFAS). Two-level hierarchy: group + subgroup.
compoundgroup	compoundno, compoundgroupno, subcompoundgroupno, convtocompoundno, convfactor	Junction table. Also holds conversion factors between related compounds (e.g. nitrate to nitrate-N).
limitlist	compoundno, consumer_max, consumer_min, waterworks_max, unit	Legal limits. consumer_max is the key field. 1,217 entries. Only 3 entries have recommended_max -- ignore it.

5. Key Reference Data

5.1 Watertype codes (drwplant.watertype)

Code	Danish	English
G	Grundvand	Groundwater (dominant -- 76,222 plants)
O	Overfladevand	Surface water (1,902 plants)
S	Soe	Lake
V	Vandløb	Stream/river
D	Draenvand	Drainage water
K	Kilde	Spring
H	Havvand	Seawater
A	Andet	Other

5.2 Key compound IDs

Compound	compoundno	Consumer limit	Unit	Notes
Nitrat	246	50	mg/l	Most common query. Real exceedance found: borehole 186.358, Frederiksvark, 56 mg/l (Mar 2026).
Nitrat-N	247	11.289	mg/l	Same as nitrat expressed as nitrogen. Conversion via compoundgroup.convfactor.
PFAS (PFOA+PFOS+PFNA+PFHxS sum)	group 110	0.002	ug/l	Strictest PFAS limit. Applies to 4 specific compounds combined.
PFAS (individual, most)	group 110	0.1	ug/l	Individual limit for most other PFAS compounds in group 110.
Sum af PFAS, 12 stoffer	group 110	0.1	ug/l	EU aggregate sum limit for 12 PFAS compounds.

5.3 Mandatory quality filters (from GEUS SQL scripts)

These filters must be applied to ALL chemistry queries. Source: official GEUS prototype SQL by bpj@geus.dk, 2022-2026.

Field	Table	Rule	Meaning
qualitycontrol	grwchemanalysis, pltchemanalysis	NOT IN (4,5,6,8,12,13,14,15)	Exclude rejected or flagged analyses.
samplestatus	grwchemsample, pltchemsample	IN (2,4,6,8,10,14) OR NULL	Sample must be authority-approved (NULL = pre-2007 reform, also valid).
project	pltchemsample	= 'DRV' OR NULL	Drinking water project code. Fixed value since 2006.
sampldate	pltchemsample	<= NOW()	Exclude 4 future-dated records (data entry errors).
attribute	both analysis tables	Check before limit comparison	NULL = normal measurement. Non-null = special: < (below DL), > (above range), ! (rejected), A (estimated). Never flag non-null as exceedance.

6. Important Gotchas & Data Quality Notes

Issue	Detail	How to handle
amount=0 is not zero	Historical records use amount=0 for below-detection measurements.	Check detectionlimit. Report as below detection if amount <= detectionlimit.
attribute field critical	Non-null attribute = special measurement. Must not trigger limit exceedance flag.	CASE WHEN a.attribute IS NOT NULL THEN NOT DETECTED before checking limits.
Unit mismatch exists	Some compounds are measured in different units than their limit value.	Check if an.unit != ll.unit. Report mismatch rather than compare incorrectly. GEUS SQL 04 documents affected compounds.
Municipality text is unreliable	borehole.municipal is free-text with pre-2007 names. Many mismatches with current municipalities.	Always use borhtownno2007 + code_808. Never the raw municipal column.
POI = contamination sites	POI tables = forureningsundersogelse (contamination investigation), not point of interest.	POI = pollution site. Relevant for contamination history queries.
Danish compound names	All compoundlist names are in Danish.	Agent must map English to Danish: nitrate = Nitrat, PFAS = Perfluorerede stoffer.
Schema prefix required	All Jupiter tables are in pcjupiterxlplusviews, not public.	All queries: pcjupiterxlplusviews.tablename
WSA coverage ends 2019	Schullehner dataset not continuously updated. Not officially authoritative.	Use w.end = 2019 for current queries. Agent should disclose this limitation.
Coordinates are UTM32	WSA polygons and drwplant both use EPSG:25832.	ST_SetSRID(ST_MakePoint(x, y), 25832). Convert user lat/lon to UTM32 before spatial queries.
drwplant includes private wells	103,195 plants includes farms and private wells, not just waterworks.	Filter with drwplantcompanytype IN (V01, V02) for public waterworks only.
Temporal intake-plant links	drwplantintake has startdate/enddate. Intakes change over time.	Filter: sampleddate BETWEEN COALESCE(startdate, sampleddate) AND COALESCE(enddate, sampleddate).
14.6M analysis rows	Unfiltered grwchemanalysis queries are very slow.	Always filter by compoundno and date range before joining other tables.

7. External Datasets

Dataset	Source	Tables	Description
WSA Polygons	Schullehner (2022). GEUS Bulletin 49. doi:10.22008/FK2/I5R1SS	pcjupiterxlplusviews.wsa	3,694 Water Supply Area polygons.

Dataset	Source	Tables	Description
			EPSG:25832. Columns: wsaid, start, end, geom.
WSA-Plant linkage	Schullehner (2022), same source	pcjupiterxlplusviews.wsa_plant	7,422 rows linking WSAID to PLANTID with temporal coverage (start_year, end_year).

Note: Cite as: Schullehner, J. (2022). Danish Water Supply Areas and their links to water production facilities: an open-access data set. *GEUS Bulletin*, 49, 8319. <https://doi.org/10.34194/geusb.v49.8319>

8. Local Setup Reference

Parameter	Value
WSL2 host	GEUS3211W10WS
PostgreSQL version	16.13 (Ubuntu 16.13-0ubuntu0.24.04.1)
PostGIS version	3.4
Database name	jupiterxl
DB user / password	pgmcp / pgmcp_dev
Schema	pcjupiterxlplusviews
Backup file	~/Thesis/data/pcjupiterxlpluspg_xl.backup
WSA shapefile	~/Thesis/data/WSA.shp (EPSG:25832, loaded via shp2pgsql)
WSA_PLANT table	~/Thesis/data/WSA_PLANT.txt (tab-separated with header)
Start PostgreSQL	sudo service postgresql start
Connect to DB	psql -h localhost -U pgmcp -d jupiterxl
Enable PostGIS (superuser)	sudo -u postgres psql -d jupiterxl -c "CREATE EXTENSION postgis;"
Restore Jupiter backup	pg_restore -h localhost -U pgmcp -d jupiterxl --no-owner --role=pgmcp -v pcjupiterxlpluspg_xl.backup

9. Next Steps

- **Priority:** Add spatial index on wsa.geom for fast point-in-polygon queries
- CREATE INDEX ON pcjupiterxlplusviews.wsa USING GIST(geom);

- **Implementation:** Adapt database.py to connect to jupiterxl with schema prefix and PostGIS support
- Build MCP tools: find_supply_plant(x, y), search_compound(name), get_water_quality(plantid, compoundno), check_pfas(plantid)
- Add geocoding: convert user address strings to UTM32 coordinates for WSA lookup
- Implement all quality filters (qualitycontrol, samplestatus, attribute, future date) in every MCP tool
- **Still to explore:** Decode remaining codes: borehole.use (find codetype via jupfield), intakeusage (codetype 816)
- Decode drwplantcompanytype values V95, V40 etc.
- Count pltchemanalysis rows and profile query performance
- Investigate pltchemanalysis column structure (note: GEUS SQL 03 uses compoundreported not compoundno -- verify column name)